SAINT MICHAEL COLLEGE OF CARAGA

VIEWS OF THE COMMUNITY AND LAW ENFORCEMENT TOWARDS EMERGENCY RESPONSE TIME: BASIS FOR THE NEED TO ESTABLISHING A CENTRALIZED EMERGENCY HOTLINE.

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

People in need can receive immediate help from crucial emergency hotlines. These hotlines enable people to directly contact emergency workers, who can quickly dispatch help to those in need. When something goes wrong, every second counts. In the event of an emergency, it is often impossible to look up hotline numbers, and the victim may not have the time to search the internet for the number (Faster Capital, 2021).

Miller and Segal (2018) and Dujua et al. (2019) examined the impact of police involvement on the growth of crime throughout a crime's life cycle. In turn, our paper examines both the effects of emergency response and the perceptions of responders and the community regarding it. Research on common emergency calls, factors that affect response times, and problems that occur during emergency responses (Capdos et al., 2023), conducted recently in Baguio City, has important implications for improving emergency response.

According to Hinata et al. (2024), both first responders and government agencies need to understand how the public responds to emergencies so that they can enhance their future communication strategies. This study aims to facilitate improvements in communication systems, resource utilization, and training programs across various industries, with a particular focus on the public safety and technology sectors. These kinds of improvements are necessary to enhance the effectiveness of emergency response systems and to improve the delivery of public services. The Davao City Local Government Unit established the Central Communications and Emergency Response Center (Davao City Central 911).

This local emergency response system operates 24 hours a day, seven days a week. This center provides a centralized and unified system that connects government resources, enabling emergency response services to be dispatched quickly, effectively, and in a coordinated manner. The Development Academy of the Philippines (2020) states that this model demonstrates how strategic coordination and communication can enhance safety for everyone.

Furthermore, people are concerned about the time it takes for help to arrive and how to communicate and distribute resources more effectively during emergenciesOttah (2025). In addition to this study, Cabral et al. (2018) note that centralized, integrated communication systems significantly reduce dispatch delays and facilitate collaboration among emergency responders.

Real-world studies show that quick police response greatly lowers the number of injuries caused by violent crimes and raises the number of arrests (DeAngelo & Weisburd, 2023; Simpson, 2024). The results of this study demonstrate the importance of seeking help promptly and arriving at the scene of an accident promptly. (Samanta et al., 2024) More proof that improvements in technology and better use of resources in centralized systems shorten response times, which backs up the push for unified emergency hotline implementations.

Additionally, the Philippine Central 911 system is a good example of how centralized emergency hotlines can be effectively utilized in real-life situations, demonstrating how the ideas behind this debate should be applied. Costa et al. (2025) state that Central 911 integrates police, fire, and medical emergency services into a single, digital platform, enabling faster routing and dispatch of calls. The Philippine National Police requires response times of three to five minutes. This centralized system has made this possible, representing a significant improvement (Philippine National Police, 2021).

The system utilizes advanced digital radio, real-time dashboards, automated incident ticketing, caller location tracking, and data integration to facilitate seamless collaboration among multiple agencies, providing a clearer understanding of ongoing operations for all parties involved. This new 911 system is the first of its kind in Southeast Asia. It offers a way to address the problems that arise with broken emergency reporting systems (Caliwan, 2025), which is truly revolutionary.

However, this next-generation 911 system is the first of its kind in Southeast Asia. It offers a revolutionary way to address the everyday problems that arise from broken emergency reporting systems. Radio operators also call for help from the right emergency personnel and ensure they stay safe while doing so. Many people associate 911 with calling for medical help, but in the United States, a significant number of 911 calls are directed to police departments (Neusteter et al., 2021). Simpson and Orosco (2021) notes that not all

callers may require emergency assistance; some calls may involve services that exceed the capabilities of more than one department.

The calls need to be answered by someone, so not having a clear request may not be a problem. These calls are not just for the 911 hotline; they are also for other emergency hotlines worldwide, such as 112 in parts of Asia and Europe, and 111 in Australia and New Zealand. People can call these numbers to contact police, fire, or medical emergency services simultaneously (Bărănescu, 2021). Problems also arise when people call 911 by accident, request more than one service, or are unsure of what they want. All 911 callers must be handled in accordance with standard operating procedures. Police call takers often deal with unclear calls, which highlights the generalist role police now play as first responders (Simpson, 2024).

Moreover, as the number of people living in cities grows, so does the need for emergency medical services that can meet prehospital care needs (Cabral et al., 2018). Injustice in the way a call taker treats a caller can make them less interested and less likely to trust, which can affect their future interactions with police (Flippin et al., 2019). This research is important because it contributes to the ongoing conversation about public safety and police work by examining different perspectives on emergency response times and establishing central hotlines.

These studies emphasize the importance of adopting a unified approach to emergency response to reduce delays and enhance efficiency. People can feel safer in their community by maintaining a strong relationship between first responders and the public (Tabbuac et al., 2021).

1.2 Review of Related Literature

This research examines various perspectives on the response time of police to emergencies and whether centralized emergency hotlines are necessary to enhance public safety and response efficiency. Tabbuac et al. (2024) study examines the time it takes for police to respond to emergencies when a central hotline is established. The speed at which the police arrive has a significant impact on the community, keeping it safe, peaceful, and orderly. It is the job of the PNP to serve and protect people not only in the community but all over the country.

In addition, the study's goal was to determine the effectiveness of police response time and its impact on crime resolution in the Municipality of Sta. St. Teresa. This study will examine ways to enhance the policing of police officers, enabling them to respond more promptly to community calls. That can help them keep good relationships with the public and even make people feel safer in their neighborhood (Tabbuac et al., 2021).

However, the distance over 7 km delays Cebu BFP arrivals to 12 minutes, allowing fires to destroy barangays before Central 911-routed crews arrive (Braga et al., 2024). Davao outer zones exceed 15 minutes amid debris (Davao City Central 911, 2024). Typhoon winds over 40 kph spread blazes faster while flooding blocks trucks in storms (Braga et al., 2024). Dalumpine (2019) notes Central 911 GPS shortens hauls, but Cardinal & Lagura (2023) Tagum studies reveal monsoon gaps worsening outcomes. Metro traffic halves engine speeds in Cebu jams (Braga et al., 2024), with OBX Solutions (2024) unified systems granting lane priority. Bargs (2024) PNP integration enhances fire dispatch, aligning with DILG (2025) rollout for weather-traffic mitigation. Pre-positioning crews addresses these core issues (Braga et al., 2024;Dalumpine, 2019).

Faster response times can improve social connections with both friends and strangers (Templeton et al., 2022). Moreover, law enforcement, especially the police, considered response as one of the key indicators of policing (Verlaan & Ruiter, 2023). For many years, police departments worldwide have recognized quick response as a crucial part of their work. Why? First, a faster response time means they are fulfilling their role as a lifeline, and second, a faster response can make more witnesses available (Verlaan & Ruiter, 202).

However, Capdos et al. (2023) found that the time it takes for police to respond to emergency calls depends on several things, such as how accurate the caller's information is, how far away, where, and how easy it is to get to the scene of the crime, the number of police officers on duty, traffic, and the availability of mobile cars. Simpson and Orosco et al. (2021), studies highlighted several factors that suggest police call classifications by dispatchers may not always align with officer assessments, which could lead to measurement error.

Researchers have written extensively about the importance of having effective emergency response systems to keep people safe and foster trust within the community. Weisburd et al. (2021) measured how police response times affect injury outcomes and

found that faster response considerably lowers injuries related to violent crime, especially for people who are already at a disadvantage.

Additionally, the importance of having systems that enable police resources to be deployed quickly to where they are needed. Did a thorough study that emphasized how centralized emergency hotlines and other integrated communication technologies make dispatching more accurate and help get resources to the scene of the emergency faster. Behavioral crisis intervention has become an important area of study in emergency response. (Murray et al., 2022) Demonstrate through qualitative interviews in Chicago that adding mental health professionals to first response teams improves the handling of behavioral emergencies.

These kinds of models aim to change the way police typically dispatch calls, and they support more specialized and diverse response plans. Shows how public safety outcomes can vary depending on how calls are reported over the phone, highlighting differences between calls to emergency (91) and non-emergency numbers. According to Simpson (2024), his research indicates that making emergency contact methods easily accessible and quick to respond is crucial for fostering trust and cooperation within the community.

The quick responses demonstrate that people are engaged with the other person in a conversation (Templeton et al., 2021). Conversations with faster response times felt more connected than those with slower response times. Even within conversations, times when people were more connected had faster response times than times when they were less connected. This effect was mainly caused by partner responsibility. It was how quickly their partner responded to them that made them feel more connected, not how quickly they responded to their partner.

One of the most important effects of the 911 system is that it works with the Philippine National Police (PNP). The PNP, led by Police General Rommel Francisco D. Marbil, has welcomed this change and understands its role as the first responder responsible for ensuring that emergency calls are answered quickly and correctly. Due to this dedication, the PNP Command Center (PCC) has a specialized call center that directs all emergency calls to the appropriate destination. The PNP has also established specialized teams of call takers and dispatchers to ensure that trained professionals handle all calls. New types of phone lines have been made available that connect national command centers to local police stations. All operations are conducted in strict adherence to the Police Operational

Procedures Manual, ensuring that responses are not only quick but also compliant with the highest standards of law enforcement (Bargs, 2024).

The Central 911 system in the Philippines shows how these ideas can be used in real life on a national level. According to the Manila Bulletin (2025), Central 911 is a single platform that the whole government uses to connect police, fire, and medical emergency services. The system quickly sends calls to the right agencies by utilizing a variety of communication technologies, including Digital Mobile Radio and Push-to-Talk cell phones.

In fact, police response times have improved significantly. Live demonstrations have shown that responses can occur as quickly as three minutes in cities, which aligns with international standards (Philippine National Police, 2021). As Capdos et al. (2023) note, this study demonstrates the complexity and multifaceted nature of emergency responses in Baguio City.

However, these cities place a high value on being close to essential services, such as hospitals, fire stations, and clinics, which makes it easier for emergency services to reach people quickly. Technologies such as IoT sensors, AI-driven analytics, self-driving drones, and real-time communication networks enable continuous data flow, providing everyone with real-time visibility into what is happening and enabling resources to be allocated proactively and facilitating the coordination of responses from multiple agencies (Costa & Westbrook, 2025).

The idea behind centralized emergency hotlines stems from resolving the conflict between the need for quick action and the fair distribution of police resources, and achieving this by leveraging technology to inform decisions based on data and enhance operational efficiency (Strom, 2017).

Understanding how first responders communicate with each other during emergencies is important for two main reasons. First, the choice of communication method can indicate how willing emergency workers are to engage with the public and how effectively they can utilize the public's knowledge to collaborate with them. For the second reason, it can show what emergency workers think about the people they are helping. A smartphone is one of the most valuable tools for dealing with natural disasters because it is carried by almost everyone at all times (Nace, 2017). However, two-way forms of communication, such as social media conversations, allow the public to help with emergencies. This may indicate that emergency workers view the public as individuals who

can offer help and ideas (Hinata et al., 202).

Direct phone lines that are always available are referred to as "hotlines." They make it easy to talk to someone right away. Marikina City is located in Metro Manila, which is prone to earthquakes and flooding due to natural disasters and human-induced problems. For this reason, the Philippine government established the 911 Hotline, which is used to assist people in trouble in the Philippines, such as during fires, natural disasters, and medical emergencies. The number 911 can also be used to stop or solve crimes across the whole archipelago (Al-Manzoor et al., 2020).

The Cebu City government, on the other hand, sought to transform the city into something akin to Singapore. A key component of this plan was establishing a central, easily accessible, and toll-free emergency hotline for the people of Cebu. The study of emergency hotlines in a city with a large population revealed major problems that made the hotlines less effective, especially during disasters and criminal cases, because people found it difficult to remember all the different numbers. This paper provides more detailed information about how the hotline is organized and operated, enabling readers to easily understand the emergency services offered by the LGU and who to contact for assistance. It also examined the arguments presented in the findings, as the community may not have been familiar with the emergency services, despite a budget being allocated for them, which could have influenced their perception. The study suggested using social media and other non-media tools to disseminate information about the emergency hotline. It also highlighted the benefits of calling 911 instead of a cell phone number, as it is free and easily accessible (Roldan et al., 2024).

This effect is mediated by how satisfied people are with the police. This study gives police departments and policymakers suggestions they can use. Abdi et al. (2023) also note that this study has both theoretical and practical implications, as it provides insights into how to enhance public perception of police institutions.

However, this article supports the creation of a unified emergency 911 system throughout the Philippines. This would make it easier for all emergency services, such as police, fire, and medical services, to respond. Law enforcement, particularly police, is continually seeking new ways to work more efficiently (Dunnet et al., 2019).

Also, Dunnet et al. (2019) put the incidents reported in the UK into four different grades. Grade 1 is the worst because it puts lives at risk or causes serious injuries. In cities,

you must respond within 15 minutes, and in rural areas, you must respond within 20 minutes. Grade 2 incidents are priority situations. They are not urgent, but they do have a time limit, like one hour for Leicester's hire police.

On the other hand, Grades 3 and 4 can be solved by making appointments. In emergencies, every second can matter, as the first person to help may be the only one who survives. When solving crimes, the same principle applies: swift action can lead to the arrest of criminals, which helps prevent crime from occurring. One of the primary objectives of the Unified 911 System is to establish a single, reliable emergency hotline for use in cases of crimes, disasters, fires, and other emergencies. According to the Manila Bulletin (2025), the implementation of the unified 911 emergency hotline nationwide could also improve the country's response to crises.

Most of people are not aware of the availability of emergency services despite sufficient budget allocation. These studies are recommend utilizing social media and other methods to raise of awareness. It also suggested that 911 is better than other mobile numbers because it is free and more memorable to recall than any other mobile number (Roldan et al., 2024).

This study compares the effects of municipal police being the first people to arrive at OHCA victims to those of non-EMS and EMS services. As the time of arrival was crucial for survival, it did not matter whether the first responder was a police officer, non-EMS personnel, or EMS personnel (Louis et al., 2023; El-Zein et al., 2023).

However, the distance between the police station and the crime scene can be a problem. According to the results of DeAngelo et al. (2023), response time decreases by 1.6% for every additional police car within 2.5 miles of the crime scene. The reason for this is that, despite the dispatcher's wishes, response times will be longer when officers are not near the scene of the crime. Nonetheless, police departments are continually seeking ways to improve their operations (Dunnet et al., 2018). Weisburd (2021) also found that police presence and quick response times directly affect reducing crime. Using detailed patrol data from the Dallas Police Department, the study demonstrated that even slight reductions in patrol presence can lead to a substantial increase in crime, underscoring the importance of maintaining sufficient and timely police deployment. Other research indicates that response effectiveness is not solely based on arrival speed, which differs from the findings of these studies. According to a study by Faroque and South (2021), law enforcement in Bangladesh

faces limited resources, inadequate coordination between agencies, and officers who are not adequately trained or aware of the problems, making it difficult for them to respond quickly and effectively to crimes.

Smart et al. (2022) also found that in Nepal, traffic police are often the first to arrive at the scene of car accidents. However, they lack sufficient training in first aid and do not have the necessary tools to assist people in need. Even when police arrived quickly, they were limited in their ability to help due to factors such as public pressure to transport victims immediately and other traffic duties. Showing how well a response works depends on more than just when it arrives. It also depends on the resources, training, and institutional support that are available.

Response differences between neighborhoods can also be attributed to variations in socioeconomic status. The paper advocates for targeted policies to mitigate these unfair situations and implement better dispatch protocols that cater to the diverse needs of communities. This illustrates the importance of allocating resources strategically (Johnson, 2021).

Additionally, as call takers, police officers in the emergency call center must deal with a variety of problems and stress. Every day, police communication workers encounter life-threatening situations and interact with individuals in distress. They must be able to make decisions in seconds (Perez et al., 2021). Some of these stressors are the need to handle life-and-death situations quickly and correctly, and call takers and dispatchers think the job has a lot at stake. The job often lacks sufficient staff and has a heavy workload, resulting in long shifts with minimal breaks and frequently requiring extra hours. Almost all of the participants reported that their job was detrimental to their health. Participants said that bad eating habits caused them to gain weight, become obese, and develop obesity-related diseases like high blood pressure and/or diabetes since starting their new jobs (Perez et al., 2021).

Similarly, 9-1-1 is the primary number to call in an emergency, and the individuals who answer the phone play a crucial role in managing responses. Due to the high volume and high level of stress, mistakes made by call handlers while collecting data can hurt the performance of 9-1-1 (Attiah, 2025). Despite the problems and stress they face, emergency medical service (EMS) workers in the US respond to approximately 37 million 911 calls each year and assist individuals who are hurt or ill.

Furthermore, family, friends, and bystanders are often the first people in the chain of survival. Creating a central place to issue operational directives and establishing a unified command structure are crucial for a successful response, particularly in avoiding jurisdictional conflicts and operational overlap between levels. During disasters, the Internal Command and Control system must be established to enable swift and accurate decision-making at all levels of command and control. There should be clear levels of authority, flexibility in roles, and a situational command approach tailored to the situation (Shafiei, 2025) within this framework. In a genuine emergency, you need to act quickly and effectively. Often, this action depends on having made and used plans to be ready. According to Cadinal and Lagura (2023), if the proper steps are not taken and the response is slow, someone could die.

Moreover, Verlaan and Ruiter (2023), via a scoping review encompassing global datasets, designate traffic congestion as the paramount predictor, with urban impediments routinely augmenting transit durations by 20-50%. This resonates with Costa and Westbrook (2025), who conceptualize "space-time compression" in 15-minute city paradigms to counteract gridlock-induced elongations. Samanta et al. (2022) scrutinize patrolling conundrums, underscoring how vehicular jams curtail spatial coverage and necessitate ad hoc redeployments. Ancillary strains manifest in dispatch operations, where Perez et al. (2021) correlate traffic-rush call influxes with dispatcher fatigue, thereby eroding call triage precision (Perez et al., 2021). In the Philippine milieu, these dynamics intensify amid Metro Manila's hyper-density, where peak-hour bottlenecks disproportionately afflict the Philippine National Police (PNP) 911 framework (Sigales, 2025; Bargs, 2024).

The response latencies predicated on incident typology, with roadway vicissitudes differentially modulating outcomes. For life-threatening cardiac arrests, El-Zein et al. (2023) and Jean Louis et al. (2023) stipulate sub-5-minute thresholds for survival optimization; congestion, however, inflates medians to 7-10 minutes in metropolitan theaters, diminishing bystander CPR efficacy. Vehicular collisions incur protracted delays of 10-15 minutes during rush epochs, attributable to on-scene inaccessibility and secondary pileups (Verma et al., 2023; Smart et al., 2022). Stationary exigencies, such as assaults or domestic disturbances, fare superior at 4-7 minutes, leveraging pedestrian ingress or peripheral routing (DeAngelo et al., 2023; Weisburd, 2021).

Futheremore, rural topographies exacerbate disparities: Svensson et al. (2024)

chronicle 11-minute averages for cardiac interventions versus 6-8 minutes for urban conflagrations, notwithstanding comparable gridlock (Svensson et al., 2024). PNP benchmarks reflect this variance, attaining 78% compliance under 5 minutes aggregate, yet vehicular urgencies skew unfavorably (Sigales, 2025; Tabbuac et al., 2024). Hsia et al. (2018) further illuminate socioeconomic gradients, wherein indigent precincts evince ambulance delays analogous to police latencies, compounding injury sequelae (Hsia et al., 2018).

5 Minutes Response Time

Sigales (2025) reports that the Philippine National Police (PNP) achieves a five-minute response time for 78.1% of 911 emergency calls nationwide, based on 5,187 calls from August 2024 to January 2025, with 4,050 responded to in 3.25 minutes on average. Bargs (2024) complements this by highlighting the revitalized 911 hotline's role in empowering PNP for faster responses, noting system upgrades that answer calls in one or two rings, as PNP Chief Gen. Rommel Marbil shifted targets from 15 to five minutes.

Furthermore, Tabbuac et al. (2024) analyzed Sta. Teresita cases, finding five-minute arrivals boost crime resolution rates and reduce backlogs through better evidence collection. Similarly, Weisburd (2021) demonstrates rapid responses enhance crime prevention by increasing police presence effectiveness.

Consequently, Cabral et al. (2018) systematic review confirms shorter times improve emergency outcomes globally, while Caliwan (2025) links PNP's speed to zero backlog goals. Moreover, Tad-awan and Capdos (2023) in Baguio showed consistent five-minute adherence raises public perception. Thus, Abdi and Hashi (2024) connect quick policing to trust and cooperation via satisfaction mediation.

Additionally, DeAngelo et al. (2023) prove faster times cut injury severity in crises. Overall, these studies affirm the five-minute standard as pivotal for PNP efficacy, public safety, and trust-building in Philippine contexts, urging sustained tech and training investments amid rising call volumes over 140,000.

Furthermore, Irwin and Pearl (2020) introduce the community responder model, which integrates multidisciplinary professionals like medics and crisis workers to respond swiftly to 911 calls that do not require armed police intervention. This model improves response times by sending the right responder quickly, reducing delays inherent in traditional systems. Svensson et al. (2024) emphasize that in rural areas, voluntary first

responders significantly improve the speed of emergency medical services during out-of-hospital cardiac arrests, underscoring the critical role of community-based rapid response in meeting stringent time goals like five minutes. Together, these studies highlight how diversifying response teams and utilizing community responders enhance the ability to meet rapid emergency response targets efficiently.

Weather

Braga et al. (2024) reveal weather hampers Cebu City's hotline responses, with rains causing slippery roads, floods, and detours pushing beyond five minutes. Shafiei et al. (2025) similarly identify storms and fog as dispatch hurdles, complicating prehospital decisions through poor visibility.

Furthermore, Roldan et al. (2024), extending Braga's work, note typhoon seasons strain Cebu operations, increasing caller risks. Hinata et al. (2024) systematic review adds that adverse weather disrupts public communication, delaying coordination. Moreover, Dalumpine (2019) describes Central 911 challenges during Philippine storms, advocating resilient infrastructure. Thus, Cardinal and Lagura (2023) in Tagum link weather knowledge to better preparedness, reducing response lags.

The broader categories incorporate natural disasters like typhoons and volcanic events, prevalent in the archipelago (Cardinal & Lagura, 2023), and man-made hazards such as industrial accidents or civil unrest (Faroque & South, 2022). Bargs (2024) emphasizes revitalized 911's role in empowering PNP for multifaceted responses, while DILG (2025) outlines the unified rollout's capacity for hybrid emergencies. These typologies, integrated with AI aids (Attiah & Kalkatawi, 2025) and training (Tabbuac et al., 2024), mitigate abusive calls (Bărănescu et al., 2021) and boost citizen satisfaction (Abdi & Hashi, 2024), fostering resilient networks amid urban growth.

Additionally, Nace (2017) highlights tech like weather apps aiding survival in natural calamities. Faroque and South (2022) connect Bangladesh floods to law enforcement strains, mirroring PNP issues. Therefore, integrating forecasts, preposition units, and drone relays as implied across studies mitigates these, ensuring reliability in typhoon-prone areas. These findings underscore weather's profound impact on PNP speed, demanding adaptive strategies for consistent service

Hêche et al. (2025) analyze external factors impacting emergency occurrences,

finding that adverse weather such as storms and heavy rain significantly delays emergency responses by impairing access and complicating dispatch coordination. These environmental constraints extend travel times and can overwhelm response systems, particularly in areas vulnerable to frequent weather disruptions.

Traffic congestion and Road condition

Traffic and road conditions represent a primary determinant of police emergency response efficacy, particularly in urbanized settings like Cagayan de Oro jams delay arrivals 4 minutes (Sigales, 2025) and also Metro Manila where density amplifies delays. Infrastructural disparities, encompassing substandard road quality and inconsistent agency resources, generate marked variances in response capabilities across services (Ottah, 2025). Verlaan and Ruiter (2023), in their comprehensive scoping review, identify traffic congestion as the predominant global predictor of delayed responses, with urban bottlenecks routinely extending travel durations. This aligns with Costa and Westbrook (2025), who advocate for 15-minute city frameworks to mitigate space-time compression challenges amid pervasive gridlock. Samanta et al. (2022) further delineate patrolling inefficiencies, where traffic jams curtail coverage and force reactive reallocations. Perez et al. (2021) highlight ancillary effects, noting that dispatcher stress from traffic-congested call volumes compromises prioritization accuracy.

However, Ottah (2025) examines infrastructural disparities causing response variances across agencies. Verlaan and Ruiter (2023) scoping review lists congestion as top predictor globally. Consequently, Costa and Westbrook (2025) advocate 15-minute city designs to compress travel amid traffic.

The response latencies exhibit pronounced disparities contingent on incident typology, exacerbated by roadway impediments. High-acuity events like out-of-hospital cardiac arrests necessitate sub-5-minute interventions to optimize survival probabilities, yet congestion inflates averages to 7-10 minutes in metropolitan contexts (El-Zein et al., 2023; Jean Louis et al., 2023). Traffic collisions incur even steeper penalties, with delays spanning 10-15 minutes during peak periods due to on-scene navigational hurdles and secondary obstructions (Verma et al., 2023). In contrast, stationary incidents such as assaults or domestics permit 4-7 minute arrivals via alternative routing or foot deployment (DeAngelo et al., 2023).

In addition, rural disparities compound these issues: Svensson et al. (2024) report 11-minute averages for cardiac arrests owing to sparse infrastructure, juxtaposed against 6-8 minutes for urban fire responses amid analogous gridlock. Philippine National Police (PNP) data reflect this, achieving 78% of calls under 5 minutes overall, though traffic skews outcomes unfavorably for vehicular emergencies (Sigales, 2025).

Hsia et al. (2018) correlate lower income locales with protracted ambulance responses, a pattern echoed in police operations where resource scarcity intersects with congestion (Weisburd, 2021). Peak-hour surges doubly burden accident dispatches, underscoring the interplay of temporal road dynamics and call complexity.

Furtheremore, Strategic countermeasures offer viable pathways to attenuate these constraints. Motorcycle patrols and traffic signal preemption diminish cardiac response intervals by 30%, enabling circumvention of immobilized thoroughfares (Jean Louis et al., 2023). Augmented dispatch systems refine routing for collisions, yielding 20% temporal reductions (Attiah & Kalkatawi, 2025). Broader urban redesigns, per Costa and Westbrook (2025), target sub-8-minute universality through proximity optimization. PNP's peak-hour protocols prioritize non-ambulant urgencies, bolstering aggregate performance (Sigales, 2025). Training regimens empower first responders to sidestep jams selectively, enhancing outcomes for time-sensitive pathologies (El-Zein et al., 2023).

Distance

Verma et al. (2023) find station distance raises crash delays 20-30%, linking to fatal outcomes. PNP NCR logs 38 remote-area stalls despite fast dispatch. Furthermore, Park and Sohn (2020) identify elderly EMS use predictors including remoteness. Neusteter et al. (2019) review 911 processing, noting geographic spreads overload systems.

Consequently, OBX Solutions (2024) pushes unified hotlines for precise routing in Philippines' sprawl. Moreover, Davao Central 911 (2024) demonstrates local mapping cuts distances. Dunnet et al. (2019) algorithms prioritize nearest units, mitigating far hauls. Thus, Micubo (2025) correlates operations with reduced crime via closer deployments.

Additionally, Strom (2017) tech impacts show GPS slashes travel. Tolito's geolocation and station optimization counter slums' circles. These strategies ensure equity, aligning distant regions with 5 minute goals amid urbanization. Cabral et al. (2018) systematic review quantifies how extra miles double mortality risks in urgencies. Dujua et al. (2019)

smartphone apps enable auto-location, slashing navigation errors by 20% in tests. Macmod et al. (2020) Marikina findings advocate satellite stations for rural equity. Thus, predictive analytics, dynamic siting, and AI routing, according to Attiah et al. (2025) bridge divides, ensuring five-minute parity nationwide despite archipelago challenges and urban boom.

El-Zein et al. (2023), the primary typologies mirror 911 service mandates under Republic Act 10639. Medical emergencies, encompassing cardiac arrests, strokes, severe injuries, and epidemics, demand paramedic prioritization, as evidenced who highlight first-responder CPR's role in out-of-hospital survival rates. Fire emergencies include structural blazes, explosions, and hazardous material incidents, often handled by the Bureau of Fire Protection (BFP), with Braga et al. (2024) documenting improved outcomes in Cebu City's hotline implementation through faster triage. Police or public safety emergencies crimes in progress, domestic violence, robberies, and disturbances form the bulk of calls, with Sigales (2025) reporting PNP's 78% response within five minutes, underscoring rapid dispatch's crime deterrence potential, Weisburd (2021).

Furthermore, Svensson et al. (2024) demonstrate that in rural settings, physical distance strongly influences response times, with community and voluntary responders playing an essential role in bridging geographical gaps to deliver timely care.

1.3 Theoretical Framework

For this study, when police respond, people often have different ideas about how long it takes them to arrive. In the context of this study, it is essential to understand how factors such as response time can influence people's perceptions of how something works. Cesare Beccaria and Jeremy Bentham developed the theory of deterrence. Fred E. Fiedler came up with the contingency theory, and Leon Festinger came up with the social comparison theory. These theories offer different but related perspectives that are crucial for understanding various aspects that influence the effectiveness of police work, the community's perception of them, and the development of policies.

Cesare Beccaria and Jeremy Bentham developed the concept of deterrence theory, which posits that punishments that are harsh, certain, and swift will be more effective in deterring people from engaging in wrongdoing. People who think they can get away with a crime are more likely to do it again (Beccaria, 1764; Bentham, 1789; Johnson, 2019).

The main idea behind this theory is that there is a connection between the severity of the punishment, its certainty, and the speed at which it will occur. According to this theory, some individuals will refrain from committing crimes if they are aware that they will be caught and face the appropriate punishment (Beccaria, 1764; Bentham, 1789; Braga et al., 2019).

This theory is related to the study because how the police react is often used to judge how well they are doing their job. This theory is a crucial component in establishing a central emergency hotline, as it demonstrates how prompt action can prevent problems from recurring (Beccaria, 1764; Bentham, 1789; Weisburd, 2021).

The study's research question, "How does the time it takes for police to respond affect the community?" is supported by this theory. For example, how safe people in the community perceive themselves to be. Bentham (1789), Beccaria (1764), and Weisburd (2021). Directing the gathering of quantitative data on the level of safety and how both the public and police officers feel about the effectiveness of centralized or unified emergency hotlines (such as 911) as a key deterrent. It also examines how having a single or centralized emergency hotline can impact the way people in the community feel during an emergency (Beccaria, 1764; Bentham, 1789; Weisburd, 2021).

Additionally, this theory focuses on how specific actions, like having a central emergency hotline or quick response times, can stop the same problems from happening again, but it does not do much to help with problems that happen by accident or because of things that were not planned (Beccaria, 1764; Bentham, 1789; McKee, 2025).

The Austrian psychologist Fred E. Fiedler developed the contingency theory in his 1964 work, "A Contingency Model of Leadership Effectiveness." It depends on the situation or context in which an organization works to find the best way to run it. Fiedler (1964) and Papagiannidis (2024) argue that there is no one-size-fits-all approach to management and that different management strategies are more effective in specific situations. This theory is related to the study of "contrasting views on law enforcement response time: the basis for the need to establish a centralized emergency hotline" because it shows that different operations cannot be handled in the same way. It also suggests that the best strategy is one that works in various situations, emphasizing the importance of being able to adapt and remain flexible. Studies also show that when the police act as a mental health crisis service, it is bad for the people who use the service, the staff, and the providers.

Also, it stresses the need for a different approach. Supporting the research questions that examined how satisfied people were with the time it took for police to respond to emergencies. It also provides a reason to establish a central emergency hotline. This theory also guides the gathering of quantitative data on how satisfied the people are with the time it takes for police to respond to emergencies in certain barangays of Butuan City. With this method, you can determine how satisfied people are with the current state of law enforcement. While the theory emphasizes that different emergencies require distinct responses, it does not provide detailed guidance on the specific strategies or approaches to employ in each case. Leaving a space for further research into the creation of specific protocols.

The Social Comparison Theory, proposed by psychologist Leon Festinger in 1954, suggests that people have an innate drive to evaluate themselves, often in comparison to others. The key concept of this theory is that people tend to assess their own attitudes, abilities, and traits by comparing them to others (Leon Festinger, 1954; Cherry, 2025).

According to DeAngelo et al. (2023) study, "Contrasting Views on Law Enforcement Emergency Response Time: Basis for the Need to Establish a Centralized Emergency Hotline." One of the factors being evaluated in terms of the police department's performance is the speed at which they respond to calls for service. Verlaan and Ruiter (2023) note that police departments worldwide strive to enhance their response times by measuring, evaluating, and publicly sharing them. They emphasize that understanding police response times is highly beneficial and can help both the police and society function more effectively.

In addition, DeAngelo et al. (2023), the fast response time of police is insufficient if it does not benefit the communities that law enforcement is supposed to protect. For fast response times are often used to measure the effectiveness of police departments in their job performance.

The theory will guide the research questions on the level of satisfaction and actual response time of the police during emergencies. Moreover, it will guide the collection of Quantitative data on the perceptions of both law enforcement officers and citizens in two selected barangays of Butuan City regarding law enforcement response times during emergencies. Using this method will provide measurable data on the level of community satisfaction with law enforcement response times. However, despite its popularity, Social Comparison Theory has been criticized for being too simplistic and for overlooking the fact

that not all individuals have a need to make comparisons to others, and that the outcomes of social comparison may depend on personal values, beliefs, and goals.

The research questions, which concern how satisfied people are with the police and how long it actually takes for them to arrive at an emergency, will be based on the theory. Additionally, it will help gather quantitative data on how people and police officers in two chosen barangays of Butuan City feel about the time it takes for the police to respond to emergencies. This method will provide us with measurable information about how satisfied people are with the speed of police response.

Social Comparison Theory is widely accepted, but it has been criticized for being overly simplistic and failing to account for the fact that not everyone needs to compare themselves to others, and that the outcomes of social comparison may depend on a person's values, beliefs, and goals.

Combining Deterrence Theory, Contingency Theory, and Social Comparison Theory provides this study on police emergency handling with a comprehensive framework. Deterrence Theory emphasizes the importance of police being able to respond quickly to emergencies. The Contingency Theory examines the reasons why different situations require different responses and how to handle various emergencies effectively. Furthermore, Social Comparison Theory clarifies why comparison can be used as a means to evaluate one's own effectiveness, such as in assessing the speed with which police respond.

In summary, these theories are the most effective in addressing the research problem because they clarify the factors that influence the time it takes for police to respond and how that time impacts the community. This theoretical framework will help the study's methodology and analysis sections flow smoothly into each other. It will also provide a strong foundation for the proposed research questions.

1.4 Schematic Diagram

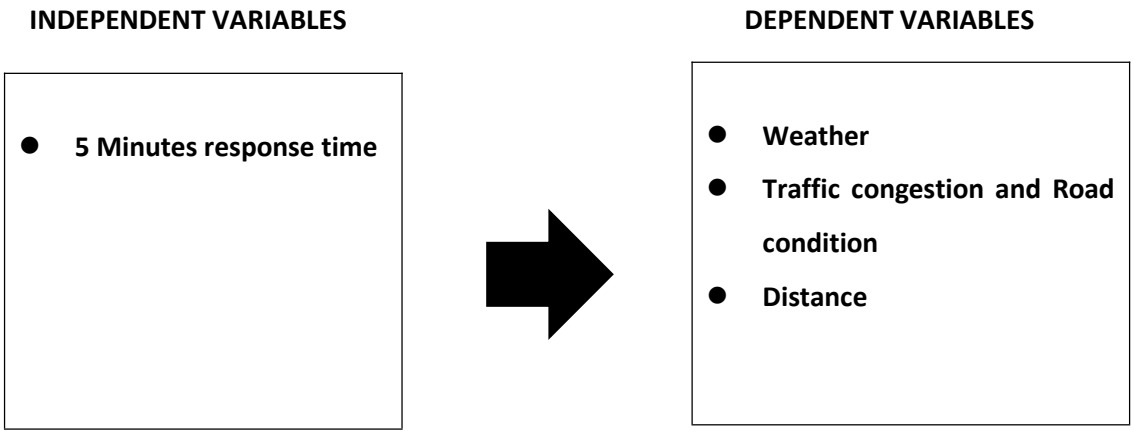


Figure 1. The schematic diagram illustrates the relationship between the dependent and independent variables of the study.

1.5 Statement of the Problem

The study aims to determine the views of the community and law enforcement towards emergency response time, focusing on the basis for establishing a centralized emergency hotline. Considering the perceived gaps between official performance and community experiences, taking into account their demographic profile. Specifically, it will seek to answer the following questions:

1. What is the level of satisfaction of community members and law-enforcement personnel regarding the current emergency response time?

   1.1. 5 minutes

2.What factors influencing emergency response time are perceived as most significant by community members and law enforcement personnel in terms of:

   2.1. Weather

   2.2. Traffic congestion and Road condition

   2.3. Distance

3.Is there a significant difference between the perceptions of community members and law-enforcement personnel concerning emergency response time?

1.6 Hypothesis

There is a significant difference between the perceptions of community members and law enforcement personnel on factors affects response time.

1.7 Scope and Limitation

This study aims to assess the views of both community and law enforcement on emergency response times in two selected barangays in Butuan City: Barangay Ampayon and Barangay Bayanihan. The findings may be generalizable to other areas during the 2025-2026 school year.

1.8 Definition of Terms

` This section defines the key terms, concepts, and variables essential to this study, providing clear and concise definitions to ensure a thorough understanding for readers.

Bystander. A person who is present during an emergency and is capable of calling emergency responders.

Centralized. Refers to a single hotline with the authority and control over any area and agencies.

Contingency. Refers to an alternative plan or solution that can be used to resolve a problem if the original plan does not work.

Crisis. A difficult, dangerous, or unstable time faced by any person during an emergency situation.

Distance. Refers to how far or near the location of the emergency is from the people responding to it.

Emergency. An unexpected event which requires specific agency.

Hotlines. Is the direct link between the user and the operator assigned to choose the specific agencies to respond during an emergency.

Law enforcement. The police agency in charge of enforcing the law, public safety, and arrest in times of crisis or emergency.

Perception. Refers to the way both the community and police view and interpret the response time of police during an emergency.

Response Time. The duration in which the police respond during emergency.

Stressful. A state of worry or mental tension on a person caused during emergency situations.

Traffic congestion. Is the condition in which road traffic becomes slow or due to an excessive number of vehicles, causing delays, longer travel time, and reduced road efficiency when responding to emergency situations.

Weather. Is the day-to-day condition of the atmosphere at a specific place and time.

CHAPTER 2

RESEARCH METHODOLOGY

This section presents the research design, research materials and methods, research environment, population and sample, research data and analysis plan, research ethical standards.

2.1 Research Design

A descriptive quantitative research design was employed to gather different perspectives on the time it takes for police to respond to emergencies. This data will be used to set up a centralized emergency response system for the 2025–2026 school year. The method enables the researcher to interpret the results in terms of theory and understand their implications for future research. It is also emphasized how important pretesting and piloting instruments are, and their goals, central areas of focus, and choice of participants are discussed. There is also discussion about data issues, such as sampling methods, determining the appropriate sample size, and maximizing the value of data and resources. There are also detailed techniques for preparing data, such as handling missing data, dealing with outliers, standardizing variables, and checking assumptions (Marc Lim, 2022).

This research employed a descriptive quantitative research design to determine whether two barangays in Butuan City, Barangay Ampayon and Barangay Bayanihan require centralized emergency hotlines. The researchers aimed to describe the features, trends, and connections in how the community perceives police emergency response times and the reasons for centralized hotlines by collecting and analyzing quantitative data. The purpose of this study was to identify patterns and correlations between these perceptions and the effectiveness of the police response. The plan involved gathering data from a large group of people through surveys or questionnaires, allowing for the study of different perspectives from citizens and police officers without personal bias influencing the results. This method provides a data-driven summary of how people perceive public safety and the effectiveness of police responses. It ensures that the research is relevant to people's experiences while providing descriptive insights into the phenomenon.

2.2 Research Materials and Methods

This study utilized a self-made questionnaire as the primary research instrument within a descriptive quantitative research design. The questionnaire has two parts. Part one collects respondent demographics. Part two examines key factors that affect emergency response time. These factors are the Philippine National Police five-minute response goal, weather, traffic, distance to site, unit availability, scene safety, resources, staffing, and communication systems

The research method employed quantitative techniques to collect and analyze numerical data related to police emergency response times. This descriptive quantitative approach enabled the researchers to comprehensively describe the perceptions of both community members and law enforcement personnel concerning the effectiveness and challenges of current emergency response operations. The numerical data allowed for detailed analysis and objective measurement of response time factors, satisfaction levels, and operational challenges faced in emergency response scenarios.

2.3 Research Locale

The study will be conducted in Butuan City, specifically in Barangay Ampayon, and in conjunction with Station 4 in Ampayon, which was established in 1990. Barangay Bayanihan, Butuan City, Police Station 3 was established in 1996. The station is a part of the law enforcement services in Butuan City, Agusan del Norte, Philippines. Butuan City is one of the highly urbanized cities in the Caraga Region. Barangay Ampayon is located along the city's main highway, while Barangay Bayanihan is a well-established community.

These barangays are selected as the research locale because of their high development and well-established communities in Butuan City, making them ideal for evaluating the views on law enforcement emergency response times. The varying access to emergency services among residents provides a significant and relevant perspective on the effectiveness and timeliness of law enforcement responses.

Before data collection, we, the researchers, requested formal permissions, considering ethical issues, to obtain potential respondents from local barangay officials and relevant authorities in Butuan City. In order to ensure ethical compliance and smooth access to participants. In addition, it will also include securing endorsements from barangay officials, which facilitates entry into communities and minimizes Potential barriers during field work.

Population and Sample

The table below illustrates the population and sample size of the gathered data from selected barangays in Butuan City, specifically Barangay Ampayon and Barangay Bayanihan, along with their respective stations.

LOCATION	POPULATION	SAMPLE	
	F	F	%
AMPAYON	14,700	278	72.0
BAYANIHAN	5,726	99	28.0
TOTAL	19,000	377	100.0

Table 1: Illustrates the total population data from Barangay Ampayon and Barangay Bayanihan, with populations of 14,700 and 5,726, respectively, along with their corresponding response rates of 72% and 28%. In total, the sample size is 19,000 and 377, with a rate of 100%.

LOCATION	POPULATION	SAMPLE	
	F	F	%
Station 3	155	92	58.9
Station 4	108	64	41.1
Total	263	156	100.0

Table 2: Shows the total population of both stations, which includes data from Station 4 in Barangay Ampayon, Butuan City, and Station 3 in Barangay Bayanihan, with populations of 155 and 108, respectively, and response rates of 58.9% and 41.1%. A total of 263 and 156 samples, with a rate of 100%

2.4 Research Data and Analysis Plan

The researchers will prepare and send a request letter to Police Station 4 and the Barangay Captain of Ampayon, Butuan City, Agusan del Norte, as well as to Police Station 3 and the Barangay Captain of Bayanihan, Butuan City, Agusan del Norte, to seek permission to conduct the study. Upon approval of the request, the researchers will employ a descriptive quantitative research design to examine the perspectives of community citizens and law-enforcement personnel from the two selected barangays in Butuan City, specifically Brgy. Ampayon and Brgy. Bayanihan.

The study will aim to collect numerical data and will utilize statistical analysis to determine and compare the perspectives of the respondents. Once permission is granted, the researchers will seek the assistance of their instructors in the conduct of the quantitative descriptive research.

After administering the survey questionnaires, the accomplished instruments will be retrieved immediately. The data that will be gathered in the study will be presented using frequency counts, percentages, and weighted mean for analysis and interpretation.

2.5 Research Ethical Standards

A.Protection of Intellectual Property Rights (IPR)

In a study on emergency response times, researchers might create original reports or surveys analyzing contrasting views. Ensuring patent, copyright, or trademark protection for these materials protects researchers' intellectual contributions, allowing secure and protected sharing of findings that advocate for centralized hotlines without risking unauthorized use.

B. Informed Consent

In this research, entitled "Views of the Community and Law Enforcement Towards Emergency Response Time: Basis for the Need to Establish Centralized Emergency Hotlines," informed consent was a significant ethical requirement. The participants were adequately informed of the purpose, aims, procedures, and importance of the research before participation. They were informed of the possible risks and benefits, and written consent

was sought to ensure that they participated voluntarily.

C.Data Privacy and Confidentiality

Research on emergency response times often involves sensitive data, including call logs and personal stories from affected individuals. To relate this ethically, researchers need to collect and protect data securely, store it with encryption, and limit access to authorized users. Building trust, especially when contrasting views reveal disparities, ensures participants' privacy when advocating for centralized systems.

D.Voluntary Participation and Freedom to Withdraw

The present research was completely voluntary. The volunteers were at liberty to choose if they would participate in the study or not. They were also made aware of their right to withdraw from the study process at any point in time without suffering any consequences, with no pressure on respondents. This ethic ensures that diverse perspectives are included without coercion, strengthening the basis for recommending centralized hotlines as a fair solution.

E.Minimization of Harm and Risk Management

The researchers made every effort to minimize potential harm or discomfort to the participants. Because of views might involve emotional topics, such as inadequate responses in crises, to minimize harm, the research should include steps to protect participants. Avoid re-traumatizing questions in interviews and offering debriefing or support resources as needed. Vitally, when discussing real-world impacts, it ensures the study promotes positive change without causing distress. Responding on a comfort level was encouraged, and all interactions were professional and respectful.

F.Beneficence and Contribution to Knowledge

This study embodies beneficence by highlighting how centralized hotlines could address contrasting views on response times, thereby maximizing benefits for society through improved public safety, equity, and harm reduction. For instance, the findings have the potential to inform policy decisions that enhance community well-being and foster goodwill among participants by making results available to them. Simultaneously, the study

advances knowledge by generating new insights into effective emergency response strategies, stimulating broader discussions, and building on existing research to drive evidence-based improvements in public services.

G. Justice and Fair Participant Selection

Participants were selected fairly based on their applicability to the study's aims, ensuring that the process was inclusive, non-discriminatory, and representative of diverse backgrounds. By prioritizing fairness in participant selection, the study captures a balanced range of perspectives on response times, which highlights potential inequalities in current systems. This approach not only upholds the ethical principle of justice but also justifies centralized hotlines as an equitable reform, promoting broader access to public safety resources and addressing disparities across communities.

H. Data Integrity and Accuracy

The report adhered to standardized measures for data collection and analysis, ensuring accuracy and reliability. It acknowledged potential biases and errors in interpretation, promoting data integrity. This approach supports valid conclusions and advocates for the establishment of centralized emergency hotlines to address disparities in law enforcement response times.

I. Transparency and Honesty in Reporting

The researchers are committed to reporting all findings honestly, accurately referencing and crediting all information, literature, and materials to prevent plagiarism and ensure intellectual integrity. This approach enhances the study's integrity and enables others to scrutinize and validate its findings, particularly in addressing disparities in public safety.

J. Use of Patented or Copyrighted Materials

The researchers conducted their study in complete adherence to intellectual property laws, ensuring that all copyrighted and patented materials, including articles, software, and digital tools, were properly cited or used with explicit permission. They avoided unauthorized use of any intellectual property while acknowledging the value of incorporating open-source materials and complying with academic and legal standards. This

ethical framework protects the study's integrity and strengthens the analysis of differing perspectives on law enforcement emergency response times, ultimately reinforcing the case for establishing centralized emergency hotlines.

K. Considerations for Human Subjects

The research maintained high ethical standards by ensuring compliance with intellectual property laws and protecting the rights and welfare of human subjects. All IP materials were properly cited or obtained with permission. At the same time, participants were treated with respect, and researchers safeguarded privacy through anonymity; all actions were aligned with approvals from Saint Michael College of Caraga. This approach upholds the study's integrity in analyzing emergency response times.

L. Responsible Use of AI and Other Technologies

Suppose researchers utilize AI tools or digital platforms responsibly for data analysis, transcription, or organization, ensuring data integrity and authenticity. Researchers consistently verified AI-generated results through human oversight, ensuring that all data was not manipulated, biased, or subjected to ethical violations. This approach reinforces the study's ethical standards and contributes to the overall reliability of the findings on law enforcement emergency response times.

M. Ethical Clearance and Institutional Approval

Before conducting the research study, researchers obtained ethics clearance and approval from the Ethics Review Committee of Saint Michael College of Caraga. The study adhered to the ethical principles upheld by the institution, the national data privacy statute, and international ethical standards in research. Ensuring that the entire research enterprise proceeded in a reliable, transparent manner and with professional and academic integrity.

References

Abdi, A. N. M., & Hashi, M. B. (2024). Impact of police effectiveness on public trust and public cooperation with the Somalia police service: exploring the mediating role of citizen satisfaction. Cogent Social Sciences, 10(1). https://doi.org/10.1080/23311886.2024.2327137

Attiah, A., & Kalkatawi, M. (2025). AI-powered smart emergency services support for 9-1-1 call handlers using textual features and SVM model for digital health optimization. Frontiers in Big Data, 8, Article 1594062. https://doi.org/10.3389/fdata.2025.1594062

Bărănescu, M., Vasilca, I. S., Tabarcia, M., Ciocoiu, N., & Nen, M. (2021). Analysis of abusive 112 emergency calls in Romania. IBIMA Business Review, 2021. https://doi.org/10.5171/2021.471463

Bargs, T. (2024). Revitalized 911 emergency hotline: Empowering PNP for a faster emergency response. Philippine National Police Journal. https://philippinenationalpolicejournal.com/revitalized-911-emergency-hotline-empowering-pnp-for-a-faster-emergency-response/

Braga, M. A., Capoy, N. M., Requierme, D., Jardin, R. A., & Roldan, M. L. R. S. (2024). Outcomes of the implementation of emergency hotline: The case of Cebu City [Unpublished manuscript]. Cebu Normal University. https://www.scribd.com/document/726778203/Bpa-4-Batch-2024-Outcomes-of-the-Implementation-of-Emergency-Hotline-the-Case-of-Cebu-City-2

Cabral, E. L. D. S., Castro, W. R. S., Florentino, D. R. de M., Viana, D. de A., da Costa Junior, J. F., de Souza, R. P., … Medeiros, A. C. (2018, December 1). Response time in the emergency services. Systematic review. Acta Cirurgica Brasileira. Sociedade Brasileira para o Desenvolvimento de Pesquisa em Cirurgia. https://doi.org/10.1590/s0102-865020180120000009

Caliwan, C. L. (2025, October 28). Napolcom aims zero case backlog by year-end. Philippine News Agency. https://www.pna.gov.ph/articles/1262029.

Cardinal, R. D., & Lagura, G. B. (2023). The influence of knowledge and practices towards emergency and disaster preparedness of Tagum City: Input in developing intervention programs. INTERNATIONAL JOURNAL OF RESEARCH AND INNOVATION IN SOCIAL SCIENCE, 7(2), 796–810.https://doi.org/10.47772/IJRISS.

Central Communications and Emergency Response Center (Central 911). (n.d.). Compendium of innovation. Development Academy of the Philippines. https://coe-psp.dap.edu.ph/compendium-innovation/central-communications-and-emergency-response-center-central-911-2/

Cherry, K. (2025). What is the social comparison process? Verywell Mind. https://www.verywellmind.com/what-is-the-social-comparison-process-2795872

Costa, D. G, & Westbrook, T. (2025). Emergency response in smart 15-minute cities: The space-time compression. Urban, Planning and Transport Research, 13(1). https://doi.org/10.1080/21650020.2025.2482831

Dalumpine, L. (2019). Central 911: Responding to people during emergencies. Philippine Information Agency. Retrieved from https://mirror.pia.gov.ph/features/2023/08/03/central-911-rsponding-to-people-during-emergencies

Davao City Central 911 Emergency Response Center. (2024). City Government of Davao. https://www.davaocity.gov.ph/departments/social-services/central-911/

DeAngelo, G., Toger, M., & Weisburd, S. (2023). POLICE RESPONSE TIME AND INJURY OUTCOMESE conomic Journal,133(654),21472177. https://doi.org/10.1093/ej/uead035

Department of the Interior and Local Government. (2025, May 26). Remulla: Unified 911 rollout to start in July 2025. Department of the Interior and Local Government. https://calabarzon.dilg.gov.ph/remulla-unified-911-rollout-to-start-in-july-2025/

Dujua, D. N. B., Garcia, G. J. V., & Sabater, H. B. B. (2019). Use of smartphone application for the selected emergency hotlines: A study on perception of residence of Mamatid and uniform personnel [Undergraduate thesis, Malayan Colleges Laguna]. ResearchGate. https://www.researchgate.net/publication/339948309_Use_of_Smartphone_Application_for_the_Selected_Emergency_Hotlines_A_Study_on_Perception_of_Residence_of_Mamatid_and_Uniform_Personnel

Dunnet, S., Leigh, J., & Jackson, L. (2019). Optimising police dispatch for incident response in real time. Journal of the Operational Research Society, 70(2), 269–279. https://doi.org/10.1080/01605682.2018.1434401

El-Zein, R. S., Kennedy, K. F., & Chan, P. S. (2023). Out-of-hospital cardiac arrest survival when CPR is initiated by first responders. Resuscitation, 190, 109914. https://doi.org/10.1016/j.resuscitation.2023.109914

Faroque, S., & South, N. (2022). Law-Enforcement Challenges, Responses and Collaborations Concerning Environmental Crimes and Harms in Bangladesh. International Journal of Offender Therapy and Comparative Criminology, 66(4), 389–406. https://doi.org/10.1177/0306624X20969938

Fastercapital. (2025). Emergency response network: Emergency preparedness as a competitive advantage: Insights for startups. November 8, 2025, from https://share.google/w65bMHnpe2AWti0EH

Flippin, M., Reisig, M. D., & Trinkner, R. (2019). The effect of procedural injustice during emergency 911 calls: a factorial vignette-based study. *Journal of Experimental Criminology*, *15*(4), 651–660. https://doi.org/10.1007/s11292-019-09369-y

Hêche, F., Schiller, P., Barakat, O., Desmettre, T., & Robert-Nicoud, S. (2025). An analytical study of external factors influencing emergency occurrences in healthcare. https://doi.org/10.1016/j.health.2025.100426

Hinata, S., Rohde, H., & Templeton, A. (2024). Communicating with the public in emergencies: A systematic review of communication approaches in emergency response. International Journal of Disaster Risk Reduction, 111, Article 104719. https://doi.org/10.1016/j.ijdrr.2024.104719

Hoonakker, P. L. T., Carayon, P., Brown, R. L., Schwei, R., Green, R. K., Rabas, M., … Werner, N. E. (2023). Satisfaction of Older Patients With Emergency Department Care: Psychometric Properties and Construct Validity of the Consumer Emergency Care Satisfaction Scale. Journal of Nursing Care Quality, 38(3), 256–263. https://doi.org/10.1097/NCQ.0000000000000694

Hsia, R. Y., Huang, D., Mann, N. C., Colwell, C., Mercer, M. P., Dai, M., & Niedzwiecki, M. J. (2018). A US National Study of the Association Between Income and Ambulance Response Time in Cardiac Arrest. JAMA Network Open, 1(7), e185202. https://doi.org/10.1001/jamanetworkopen.2018.5202

Irwin, A., & Pearl, B. (2020, October 27). The community responder model. Center for American Progress. https://www.americanprogress.org/article/communityresponder-model/

Jean Louis, C., Cildoz, M., Echarri, A., Beaumont, C., Mallor, F., Greif, R., … Reyero, D. (2023). Police as first responders improve out-of-hospital cardiac arrest survival. BMC Emergency Medicine, 23(1). https://doi.org/10.1186/s12873-023-00876-w

Johnson, R. (2021). Early origins of racial health disparities: Human capital policy is health policy. Health Affairs, 40(3), 400-408. https://doi.org/10.1377/hlthaff.2020.01117

Lee, D. S., & McCrary, J. (2017). The deterrence effect of prison: Dynamic theory and evidence. *Advances in Econometrics*, *38*, 73–146. https://doi.org/10.1108/S0731-905320170000038005

Macmod, M. A., Vargas, K. A. A., Umipig, E. M., & Tella-in, M. C. J. (2020). 911 Marikina 2020. https://www.scribd.com/document/446870108/911-Marikina-12-18-UPDATED-docx

Manila Bulletin. (2025, May 31). A unified 911 system that citizens can rely on. Manila Bulletin. https://mb.com.ph/2025/05/31/a-unified-911-system-that-citizens-can-rely-on

McKee, Z. (2025, November 5). Reducing violence for a SAFER future: Enhanced prevention of violence against women and girls. Police Chief Online. Retrieved from. https://www.policechiefmagazine.org/reducing-violence-safer-future/?ref=89e73a190f5830a3e7e9ac2e83ce0a9f

Micubo, H. G. (2025). Implementation of police operations in relation to crime rate. EPRA International Journal of Multidisciplinary Research (IJMR), 11(4), 721-729. https://doi.org/10.36713/epra21249

Miller, A. R., & Segal, C. (2018). Do Female Officers Improve Law Enforcement Quality? Effects on Crime Reporting and Domestic Violence. Review of Economic Studies, 86(5), 2220–2247. https://doi.org/10.1093/restud/rdy051

Murray, C. H., Contreras, J. L., Kelly, C. H., Padgett, D. K., & Pollack, H. A. (2023). Behavioral Crisis and First Response: Qualitative Interviews with Chicago Stakeholders. *Community Mental Health Journal*, *59*(1), 77–84. https://doi.org/10.1007/s10597-022-00990-2

Nace, T. (2017, December 15). How technology is advancing emergency response and survival during natural disasters.Forbes. https://www.forbes.com/sites/trevornace/2017/12/15/how-technology-is-advancing-emergency-response-and-survival-during-natural-disasters/

Neusteter, S. R., Mapolski, M., Khogali, M., & O'toole, M. (2019). The 911 Call Processing System: A Review of the Literature as it Relates to Policing From the Director. Vera Institute Of Justice, (July).

OBX Solutions Technology Inc. (2024). Enhancing emergency response efficiency through a unified hotline system in the Philippines. Retrieved from https://obxsolution.com/

Ottah, Alice N., "Socio-economic and Infrastructural Determinants of Response Time Disparities across Emergency Service Agencies" (2025). Electronic Theses & Dissertations (2024 present). 192. https://scholarsarchive.library.albany.edu/etd/192

Papagiannidis, S. (Ed.). (2024). TheoryHub: Academic theories reviews for research and teaching and learning. Newcastle University Business School. https://open.ncl.ac.uk

Park, J. M., & Sohn, A. (2020). Predictors affecting the elderly's use of emergency medical services. Osong Public Health and Research Perspectives, 11(4), 209–215. https://doi.org/10.24171/j.phrp.2020.11.4.10

Perez, R. A., Jetelina, K. K., & Reingle Gonzalez, J. M. (2021). The Chronic Health Effects of Work-Related Stressors Experienced by Police Communications Workers. Safety and Health at Work, 12(3), 365–369. https://doi.org/10.1016/j.shaw.2021.05.005

Directorate for Personnel and Records Management, Philippine National Police. (2025). DPRM Citizen's Charter 2025 (2nd ed.) [PDF]. Philippine National Police. https://dprm.pnp.gov.ph/wp-content/uploads/2025/05/DPRM-Citizens-Charter-2025-2nd-edition_1.pdf.

Redentor D. Cardinal, Glenne B. Lagura DPA (2023). The Influence of Knowledge and Practices Towards Emergency and Disaster Preparedness of Tagum City: Input in Developing Intervention Programs. International Journal of Research and Innovation in Social Science (IJRISS), (2023), 796-810. https://doi.org/

Roldan, M., Braga, M. A., Capoy, N. M., Requierme, D., & Jardin, R. A. (2024). Outcomes of the implementation of emergency hotline: The case of Cebu City. Bachelor of Public Administration, Department of Public Governance, Cebu Normal University. https://www.scribd.com/document/726778203/Bpa-4-Batch-2024-Outcomes-of-the-Implementation-of-Emergency-Hotline-the-Case-of-Cebu-City-2

Samanta, S., Sen, G., & Ghosh, S. K. (2022). A literature review on police patrolling problems. Annals of Operations Research,316(2),10631106. https://ideas.repec.org/a/spr/annopr/v316y2022i2d10.1007_s10479-021-04167-0.html

Shafiei, M., Tavan, A., Moradi, S.M. et al. Multidimensional determinants of disaster response in prehospital emergency dispatch centers: content analysis of expert perspectives. BMC Emerg Med 25, 208 (2025). https://doi.org/10.1186/s12873-025-01354-1

Shannon, C. (2017). Understanding community-level disaster and emergency response preparedness. Disaster Medicine and Public Health Preparedness, 9(3), 239–244. https://doi.org/10.1017/dmp.2017.28

Svensson, A., Nilsson, B., Lantz, E., Bremer, A., Årestedt, K., & Israelsson, J. (2024). Response times in rural areas for emergency medical services, fire and rescue services and voluntary first responders during out-of-hospital cardiac arrests. Resuscitation Plus, 17, Article 100548. https://doi.org/10.1016/j.resplu.2023.100548

Sigales, J. (2025, January 13). Police respond to 78% of 911 calls within five minutes – PNP. INQUIRER.net. https://newsinfo.inquirer.net/2024568/police-respond-to-78-of-911-calls-within-five-minutes-pnp

Simpson, R. (2024). Calling the Police via 911 Versus a Non-Emergency Number: Variation Among Telephone Reporting Methods and their Implications for Public Safety. Crime & Delinquency, 0(0). https://doi.org/10.1177/00111287241295679

Simpson, R., & Orosco, C. (2021). Re-assessing measurement error in police calls for service: Classifications of events by dispatchers and officers. PLoS ONE, 16(12 December). https://doi.org/10.1371/journal.pone.0260365

Smart, G., Banstola, A., Raut, R., Ghimire, K., Mytton, J., Joshi, E., & Joshi, S. (2022). Post-Crash First Response by Traffic Police in Nepal: A Feasibility Study. *International Journal of Environmental Research and Public Health*, *19*(14). https://doi.org/10.3390/ijerph19148481

Strom, K. (2017).Research on the impact of technology on policing strategy in the 21st century: Final report (Document No. 251140). National Institute of Justice. Award No. 2012-MU-CX-004.U.S. Department of Justice. https://www.ojp.gov/pdffiles1/nij/grants/251140.pdf

Tabbuac, J. T., Asuncion, J. A., Javier, H., Torrate, J., & Balagan, J. (2024). Analyzing the effectiveness of police response time on crime resolution in the Municipality of Sta. Teresita: A case study .Cagayan State University. https://www.studocu.com/ph/document/cagayan-state-university/bs-criminology/case-study-pnp-sta-please-read/109508430

Tad-awan, F. P., & Capdos, M. J. (2023, May 5). Baguio City Police Office response to emergency calls (SSRN Scholarly Paper ID 5290027). SSRN. https://doi.org/10.2139/ssrn.5290027

Tanudra, J., Tañega, A., & Las, A. (2017). Public perception on the quality of response of police officer stocrime incidents. http://103.123.43.47:8080/handle/20.500.14045/400

Templeton, E. M., Chang, L. J., Reynolds, E. A., LeBeaumont, M. D. C., & Wheatley, T. (2022). Fast response times signal social connection in conversation. Proceedings of the National Academy of Sciences of the United States of America, 119(4). https://doi.org/10.1073/pnas.2116915119

Wang, Y., Liu, H., Li, W., Li, H., Zhu, S., & Pan, Z. (2023). Analysis on the evolution path of Chinese subjective attitude a prediction model based on structural equation (p. 37). SPIE-Intl Soc Optical Eng. https://doi.org/10.1117/12.2679173

Weisburd, S. (2021). Police presence, rapid response rates, and crime prevention. Review of Economics and Statistics, 103(2), 280–293. https://doi.org/10.1162/rest_a_00889

Verlaan, T., & Ruiter, S. (2023, December 1). Predictors of police response time: a scoping review. Crime Science. BioMed Central Ltd. https://doi.org/10.1186/s40163-023-00194-3

Verma, S., Wilson, F., Wang, H., Smith, L., & Tak, H. J. (2023). Impact of Community Socioeconomic Characteristics on Emergency Medical Service Delays in Responding to Fatal Vehicle Crashes. AJPM Focus, 2(4). https://doi.org/10.1016/j.focus.2023.100129